

Flu Forecasting Project DRAFT

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Abstract

Each year, influenza epidemics result in significant morbidity, mortality, and cause adverse effects to the economy. Predicting annual influenza outbreaks can prepare jurisdictions to reduce detrimental outcomes through targeted vaccinations, health care preparations, and other programs to reduce exposure. Since probability of exposure among individuals is enhanced through increased human density, we used incidence data from the Southern hemisphere during the winter to predict future influenza rates in the Northern hemisphere. The model predicted a 30% increase in the winter of 2025 in North America which closely tracked actual incidence. We maintain models populated with data from other jurisdictions can be useful to predict relative incidence and be used to more effectively allocate resources from which to protect public health.

1 Introduction

Annual influenza outbreaks cause considerable morbidity, mortality, and economic effects. There is also widespread concern amongst public health professionals regarding the deadly potential of a future influenza pandemic (Doshi 2008). Pathogenic airborne viruses increase transmission proportionally within higher areas of human density. Such increases occur annually during winter in temperate regions when people spend greater time indoors (Simonsen 1999). Tempo-spatial patterns can be useful in predicting influenza outbreaks (Onozuka and Akihito 2008, Shaman and Karspeck 2012) and can be used in modeling approaches to better prepare certain jurisdictions to focus public health resources.

Motivate need for long range forecasts, drug suppliers, vaccine influence etc.

Current state of the art forecasts are done using an ensemble of models from teams in CDC, but produce forecasts in 1-4 weeks.

The strength of an influenza outbreak can be dependent on a variety of factors such as the dominant strain(s) of the virus $H(i)N(j)$, efficacy of the vaccine itself, and proportion of population that received a timely vaccination. Vaccine efficacy in the U.S. is related to how accurately one can forecast the dominant viral strains likely to infect humans and has had a relatively small variance in the past 10 years with a 36(9)% reduction in infection.

Rather than studying the virus evolution itself one can use the southern hemisphere as a testing ground for the upcoming season.

Here we present a model designed to predict future influenza incidence in the Northern hemisphere using data collected during the winter season from the Southern hemisphere.

2 Southern Hemisphere Countries

The nature of the Earth's tilt, with respect to its orbit around the Sun, provides a phase difference for the seasons of northern and southern hemisphere locations making winters $\sim \frac{1}{2}$ years apart. Given

the ease one can travel the globe, it is reasonable to assume there exist countries in the southern hemisphere that can provide a picture of what we can expect in the US.

We select a list of relevant “candidates” by borrowing influenza records from the WHO and compute their lagged correlations with the U.S. hospitalizations records given by the CDC (specifically this is the flu-surv dataset which represents $\sim 10\%$ of the population)¹.

Specifically we compute

$$C_{\alpha}(\delta t) = \frac{\sum_t (x_t^{US} - \bar{x}^{US})(x_{t-\delta t}^{\alpha} - \bar{x}^{\alpha})}{\sqrt{\sum_t (x_t^{US} - \bar{x}^{US})^2 \sum_t (x_{t-\delta t}^{\alpha} - \bar{x}^{\alpha})^2}} \quad (1)$$

where x_t^{α} represents the number of positive cases reported for country α on week t .

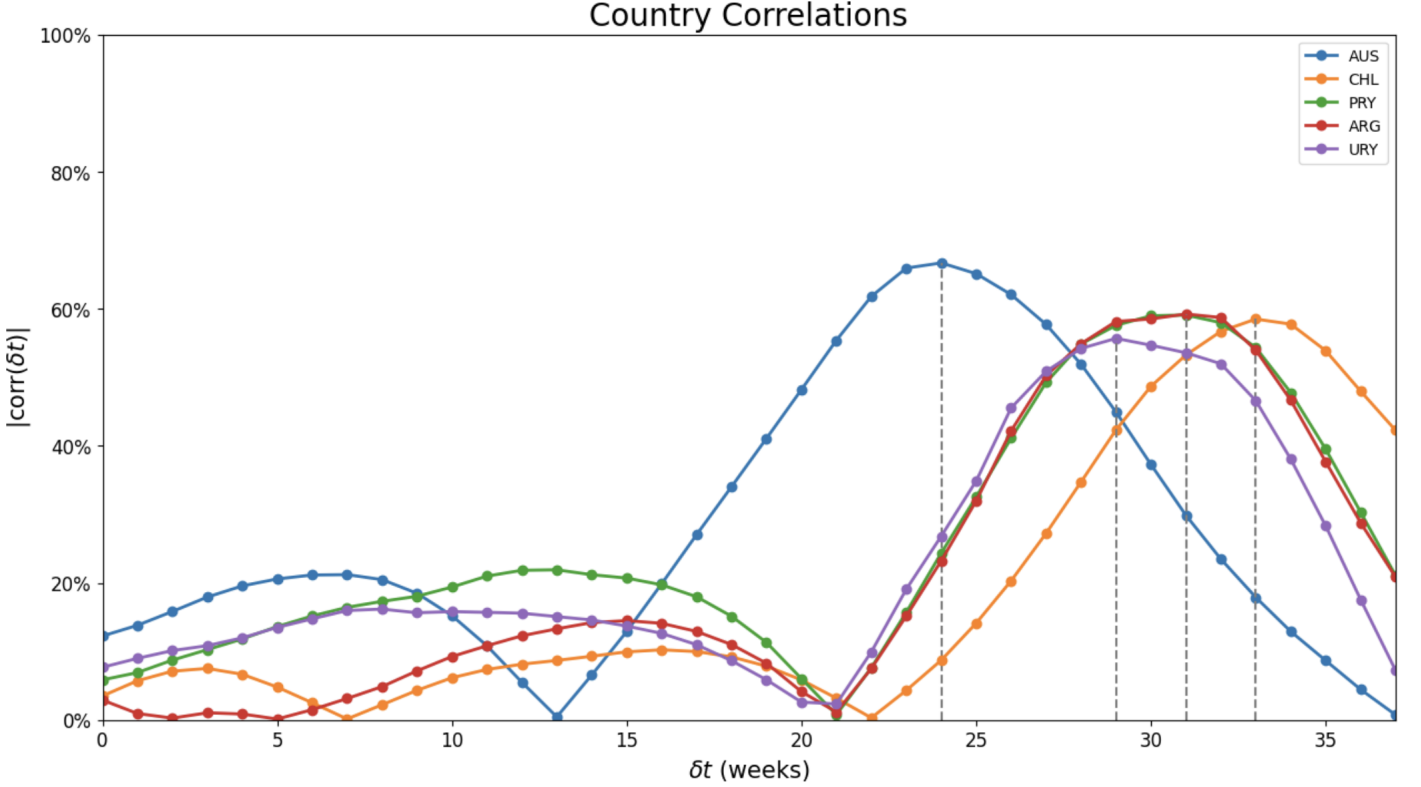


Figure 1: The absolute value of correlations between a select few southern hemisphere country’s influenza case records and the US total influenza hospitalization rates. The correlations are with respect to the number of lagged previous weeks (δt).

As seen in Figure 2, we find the countries Australia, Chile, Paraguay, Argentina, and Uruguay have the clearest signal in our target window of $\delta t \geq 20$. These countries have clear peaks and allow us at least a 20 week forecast from a given week.

We can see in Figure 2 that there is no single candidate or obvious linear combination of the group that could represent the target. Each candidate has its own peak height and location. There may be a non-linear relationship that can be captured via tuning neural network. We now move to creating models that can make an improved prediction than just examining southern countries by eye.

¹We could just as well have used WHO data as the target as well, but CDC hospitalizations were chosen to avoid bias from having the entire project built on a single data source.

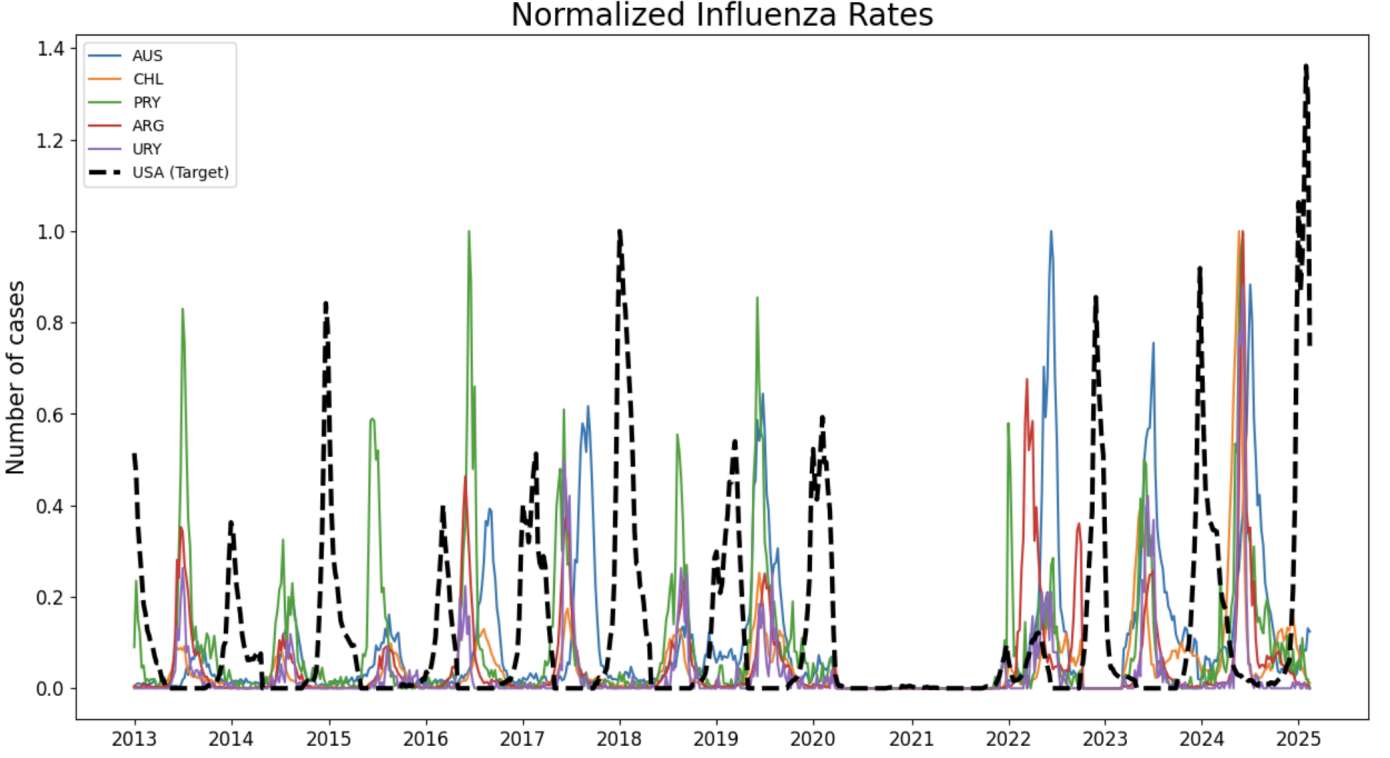


Figure 2: The US hospitalizations (black, dashed) between 2015 and the beginning of 2023 compared to the 5 southern hemisphere countries total influenza cases documented by WHO. All countries are normalized such that 1 reflects the most severe week that country experienced as documented by the WHO and the CDC.

3 Models

Mention efforts for attempting linear relationships with lagged features. Complain about seasonality with weeks.

RNNs were once the state of the art for sequence learning commonly used in things like time-series forecasting and text sequence generation. These are now largely replaced by CNNs and transformer models. An advantage for small datasets could be RNNs.

We employ RNNs to utilize feedback connections that enable them to learn long sequences of data where ordering is important. These have the particular advantage of not needing a parametric description to capture the seasonality. The input features the 5 southern countries and the US influenza cases in batches of 26 week long sequences shifted by 20 weeks. Each country is normalized such that the maximum rate in the time window 2013-2021 is 1.

The architecture of the models follows the general recipe

$$o^t = \mathbf{B}_{\mu'\nu'} \mathbf{A}_{\mu'\nu',\mu\nu} \mathbf{i}_{\mu\nu}^t \quad (2)$$

where the separable pieces $\mathbf{A}, \mathbf{B}$ are defined as

$$\mathbf{A}_{\mu'\nu',\mu\nu} \equiv \mathbf{A}_{\mu'_N \nu'_N, \mu_1 \nu_1} = \left(\prod_{n=1}^N \mathbf{L}_{\nu'_n \nu''_n}^{(n)} \sigma_{\mu'_n \nu''_n, \mu''_n \nu_n}^{(n)} \mathbf{R}_{\mu''_n \mu_n}^{(n)} \right) \quad (3)$$

$$\mathbf{B}_{\mu\nu} = \mathbf{L}_{\gamma'''_N} \left(\prod_{n=1}^N \mathbf{L}_{\gamma'''_n \gamma''_n}^{(n)} \mathbf{D}(p)_{\gamma''_n \gamma'_n}^{(n)} \sigma_{\gamma'_n, \gamma_n}^{(n)} \right) \mathbf{F}_{\mu \cdot \nu \equiv \gamma} \quad (4)$$

where $\mathbf{L}_{\gamma\gamma'}$ is a linear layer of shape $[\gamma, \gamma']$ (if only one index is included than it is assumed that $\gamma = 1$), σ is the activation function (generally RELU unless otherwise stated), $\mathbf{D}(p)$ is a dropout

tensor with probability p , $\mathbf{F}$ is a flattening tensor, and $\mathbf{R}$ is the RNN where we make use of the slight different flavors of GRUs, LSTMs, and conventional RNN. The model architecture is visually represented in figure 3

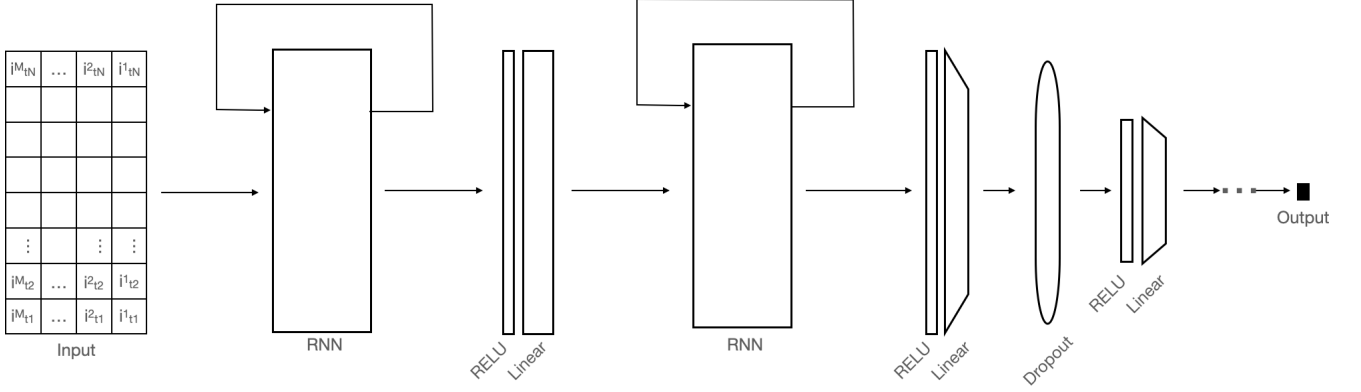


Figure 3: Model archetype composed of two varying combinations of RNNs and sequential layers.

The **A** piece of the model is inspired by the auto-encoder RNN design (cite here), with the intention of the first layer to encode the relevant features from the input from the candidates and the second RNN layer to act as the decoder. We include The **B** portion is designed to find the possible non-linear combinations of the individual countries that may best describe the U.S. flu season, as well as controlling over-fitting via the drop-out layers. This formulation allows for infinite model architectures to choose from and we explore a select few combinations creating an ensemble of models.

This is an input tensor of shape $[T_{window}, 26, 6]$ where T_{window} is the total weeks included in the fitting procedure. The input tensor has a 21-week lag in the time-sequenced index, so for a given week t , there will be $t - 20, t - 21, \dots, t - (20 + 26)$ sequenced data. This shift allows us to predict a data point that is 21 weeks away from the time the data is collected, and the sequence of 26 weeks is chosen such that is a sufficiently long enough window for the RNN to learn the pattern. Due to the nature of the normalization, insignificant pieces of the tensor will be nearly zero and will be suppressed compared to the peaks in the gradient when updating parameters.

Covid input was replaced with zeros.

Paper attempts to predict both of the 23-24 and 24-25 seasons using the previous year's season as validation.

Unrealistic to think a model with such a long forecast and no input of United States feature can capture week-to-week behavior.

Create an adjusted training metric that aims to capture most importantly the peak height, secondly location.

Training metric is

$$MSE^{adj} = \sqrt{\left(y_{val}^{max} - y_{val}^{pred,max}\right)^2 + \alpha^2 \left(\left(t_{val}^{max} - t_{val}^{pred,max}\right)^2 + MSE_{train}\right)} \quad (5)$$

Training metric used to choose model is a hybrid of training MSE and how well the peak height and location are described in the validation season. The training MSE is included to ensure the model has trained sufficiently but is suppressed by a factor of $\alpha = 0.1$ so that the peak height is still the dominant feature when searching for a minima or plateau in the training curve.

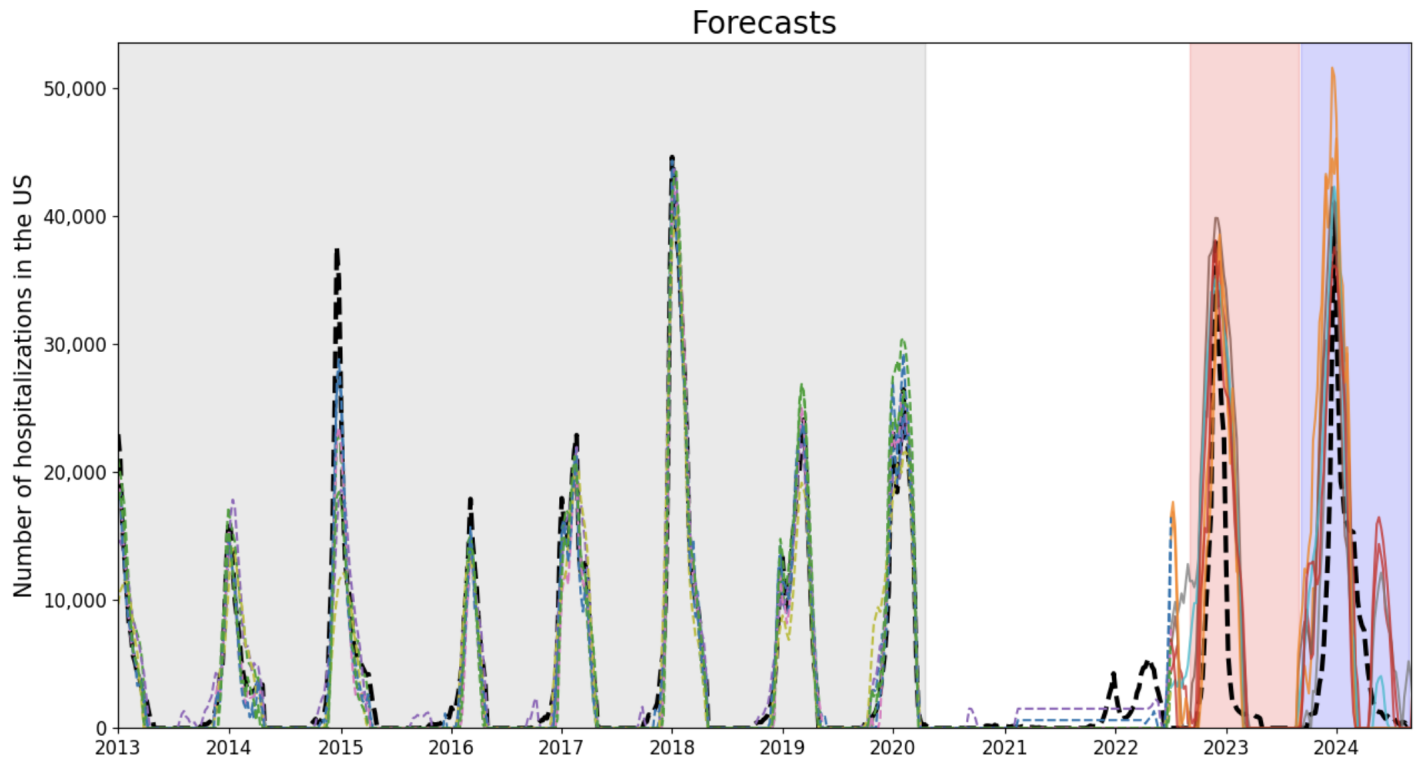


Figure 4: Dashed line represents the target U.S. hospitalization rate. Different colors represent different accepted models with the dashed in the training region and solid lines are outside. Red shaded region includes the validation region, and blue shaded represents the test region.

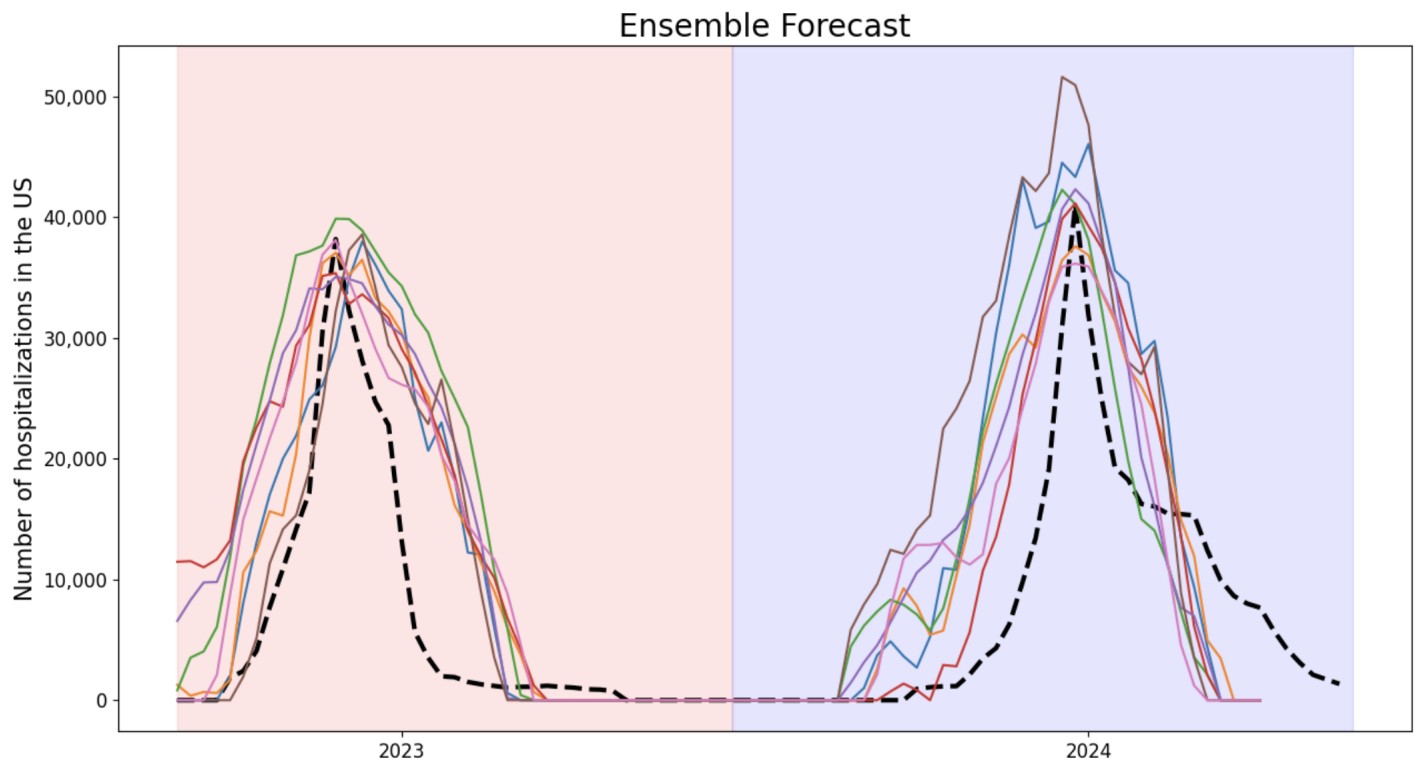


Figure 5:

A simple average of the model variation will produce a poor description of the data due to the

seasonality. One must first capture the phase differences and shift accordingly. A parametric solution to combining is the following sum of renormalized hybrid Lorentzian distribution

$$L(t) = \sum_i A_i \frac{\left(\frac{\Gamma_i}{2}\right)^2}{(t - t_i)^2 + \left(\frac{\Gamma_i}{2}\right)^2} \quad (6)$$

where A_i is the peak height, t_i peak location, and Γ_i is the half width. A typical Lorentzian has a defined normalization that we have removed to control the height. All three of the parameters can be computed for each model and their averages and deviations can be used to estimate the model variance. Use a sum due to the double peak nature of some seasons.

Below figure demonstrates these parameterizations work remarkably well.

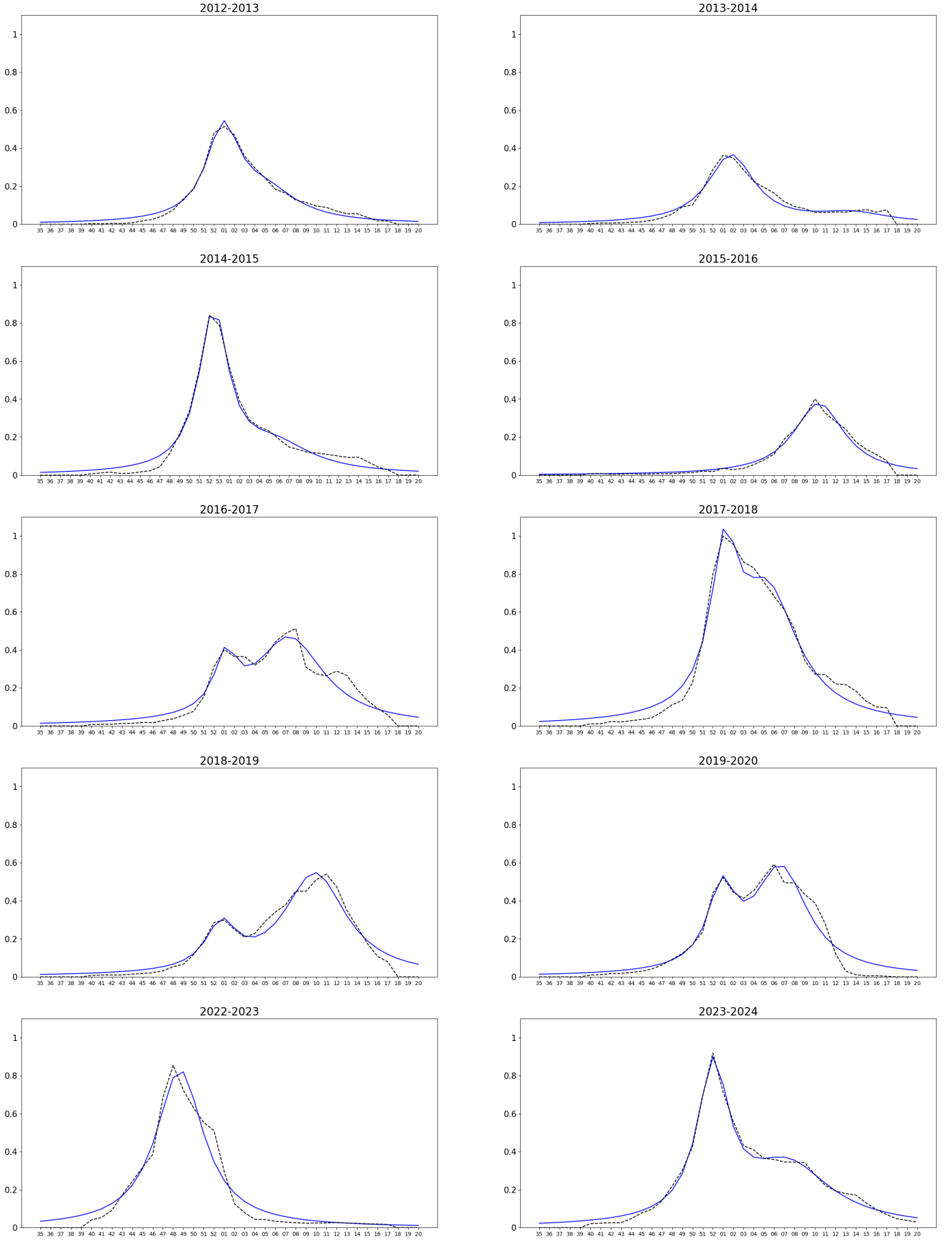


Figure 6: Double Peak fits.

Make a single peak approximation due to the sensitivity of the data.

We note the validation model has over-predicted the width by roughly a factor of 2 which we can account for in the test piece.

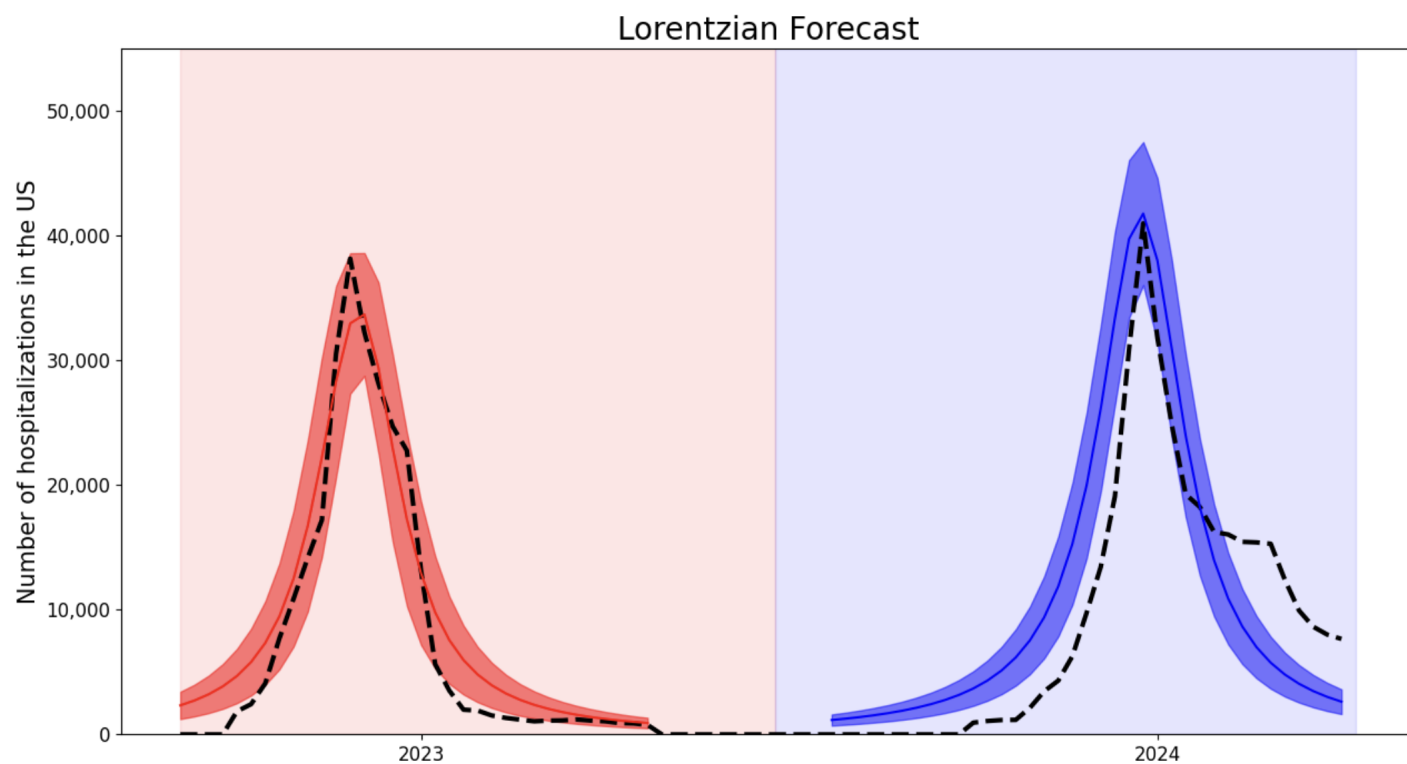


Figure 7:

Repeat the same procedures for next season.

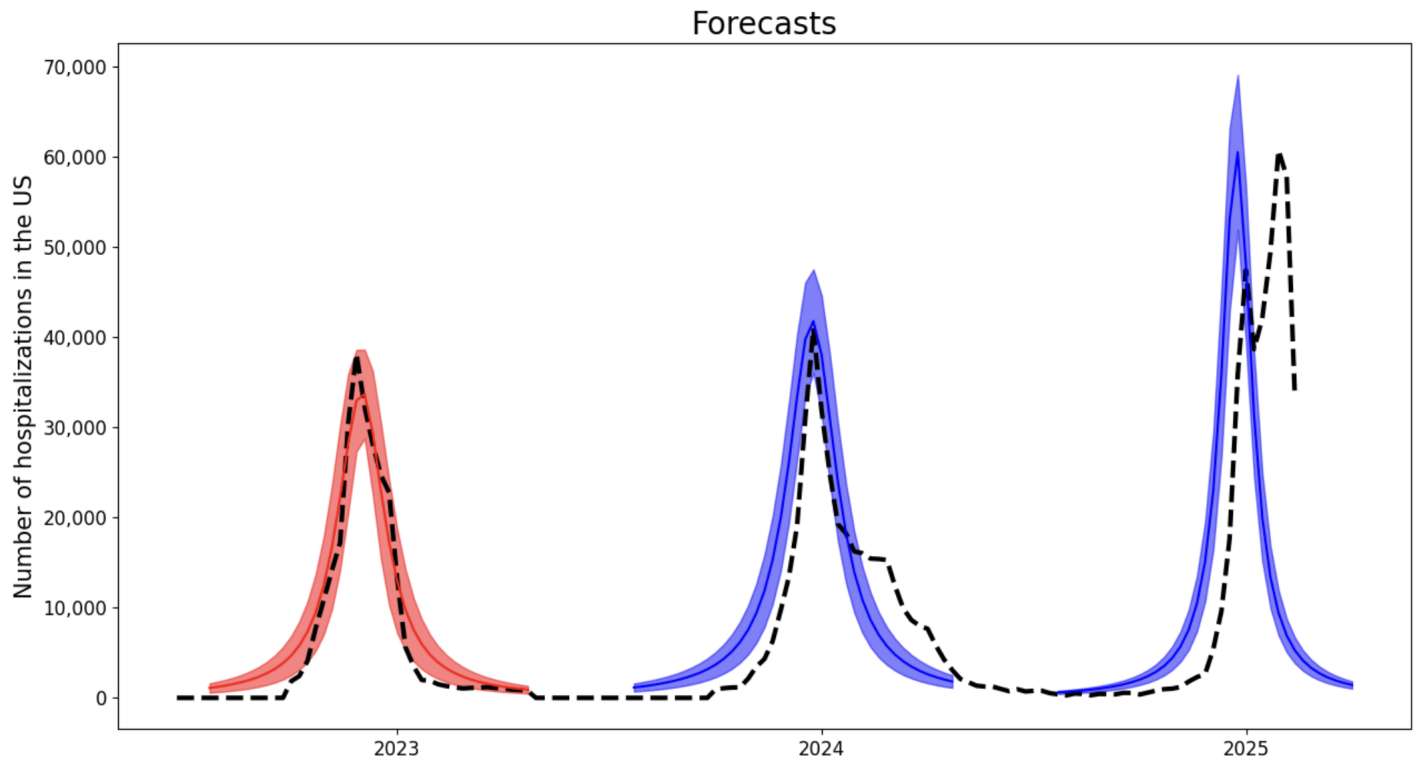


Figure 8:

Remark one could in principle use fit parameters as targets and reduce the dimensionality of the problem, however training becomes challenging because each year is reduced to a single data point. Note factors that influence multi peak behavior.

Captures dominant peak behavior but it appears to produce same peak location as previous season. Description also lacks the ability to capture the double peak behavior the flu season experiences.

4 Conclusion